

TXOne Networks

2022
/Q3

OT Zero Trust

Drives Automobile Quality

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Automakers rely heavily on automation. There can be as many as 700 robots needed to assemble one car.[1] Large industrial robots move vehicles along the assembly line, sometimes lifting them into the air so every inch of space in the factory is used efficiently. Giant arms spot weld doors, panels, and other large auto parts. Smaller robots weld mounts and brackets. Collaborative robots called "cobots" assist humans with installing car seats and other tasks that require the human touch. They add pinpoint precision control and consistency to augment human abilities.[2] Endpoints on the shop floor range from computer tablets to robots and cobots to human machine interfaces (HMI). Computer networks ferry messages, much like a superhighway, for routing control commands and other important messages throughout the plant from endpoint to endpoint.

Automakers produce vehicles that can have more than 100 sophisticated electronic control units onboard. ECUs receive data from environment sensors that relay up-to-minute readings about temperature, road conditions, and vehicle health.[3] They also provide infotainment, GPS, and other features. All in all, automobile manufacturers face many attack surfaces.[4] Cybersecurity has become the new marker of quality for vehicle design and manufacture. OT zero trust drives quality for automakers with the four cornerstones of cyber defense: ***inspect, segment, lockdown, and reinforce***.



[1] *Toyota Production Documentary Toyota Manufacturing Production and Assembly at Toyota Factory. Video at https://www.bing.com/videos/search?q=Toyota+Production+System+Documentary+-+Toyota+Manufacturing%2c+Production+and+Assembly+at+Toyota+Factory&docid=607989879896485294&mid=1FADF1767254A10A9FCC1FADF1767254A10A9FCC&view=detail&FORM=VIRE (accessed May 17, 2022).*

[2] *How Are Robots Used in Car Manufacturing?, https://arminstitute.org/news/robots-in-car-manufacturing/ (access May 17, 2022).*

[3] *What is an Electronic Control Unit (ECU)? (autopi:2021), https://www.autopi.io/blog/what-is-electronic-control-unit-definition/ (accessed May 17, 2022).*

[4] *Ondrej Burkacky, Johannes Deichmann, Benjamin Klein, Klaus Pototzky, and Gundbert Scherf, Cybersecurity in automotive, (McKinsey & Company:2020), https://www.mckinsey.com/~/media/mckinsey/industries/automotive%20and%20assembly/our%20insights/cybersecurity%20in%20automotive%20mastering%20the%20challenge/cybersecurity-in-automotive-mastering-the-challenge.pdf (accessed May 17, 2022).*

Cyber Attacks on Automakers



Malware inside an automotive manufacturing plant may literally or figuratively "pop the clutch" and stall your assembly line - or worse yet, it could cause a robot to go rogue. Ransomware recently halted production at a few global automakers. At one such plant, an ex-employee attempted an insider breach. Unsecured memory at a cloud provider resulted in exposed passwords, API keys, and tokens. These are some of the attacks that automobile manufacturers must be ready to face.

WannaCry Ransomware Attack

In May 2017, another automaker was hit by a worldwide ransomware attack that affected production.[5] This time, the ransomware was the WannaCry worm. This cyber attack halted or reduced the output of at least five production plants. The automaker declined to comment on their cybersecurity policies because they "would be sending a message to potential hackers" about how to hack them in the future.[6] That's because WannaCry primarily impacts organizations using legacy versions of Windows that are no longer supported and whose firewalls do not filter traffic. This old-fashioned type of cybersecurity-by-obscurity is no longer a viable option. The OT zero trust portable security scanner was designed to protect legacy systems just like these, along with air-gapped devices.

Don't Trust – Don't Cry

WannaCry cannot attack legacy Windows systems when OT zero trust is protecting them. In the past, it was difficult to protect legacy equipment, but within the OT zero trust framework, a portable security device is essential for inspecting every air-gapped or stand-alone devices for malware. This security inspection also takes a detailed snapshot of asset data that includes computer information, Windows Update status, and application lists while simultaneously updating your inventory. Asset data can be collected from legacy equipment such as stand-alone PCs, even those running Windows XP, Windows 7 or Linux. Air-gapped systems that were previously undetectable by scans can also be inspected. During the normal production lifecycle, you can schedule routine inspections or conduct surprise checks. These inspections do not interrupt device performance, since there is no software installation required for security checks or removal of malicious code. Scanning tools run on a portable tool resembling a USB flash drive.

[5] *"Nissan's Sunderland plant hit by cyber-attack", BBC News (2017) https://www.bbc.com/news/uk-england-39906534 (accessed May 17, 2022).*

[6] *"Laurence Frost and Naomi Tajitsu, "Renault-Nissan is resuming production after a global cyberattack caused stoppages at 5 plants", Business Insider (2017) https://www.businessinsider.com/renault-nissan-production-halt-wannacry-ransomeware-attack-2017-5 (accessed May 17, 2022).*

Asset information can be exported from the Portable Security Management Program into machine-readable .CSV files so you can see an integrated view of scans occurring across multiple scanning tools and endpoints. Pattern file updates, configuration deployment, correlation of scan logs, and asset information can all be performed through this program to meet administrator policy requirements.

Easily Secure Air-Gapped Systems

Safeguarding physical security is easy with a portable security device. First, plug the portable security device into the Defense Console or another computer with secure internet access. Download the latest virus signature-pattern files from the server. Configure scan settings and specify the folders to be scanned, along with what actions the tool should take after detecting a threat. This plug-and-play device will perform virus scans or audits and collect system information. LEDs on the portable security device turn red when malware has been detected and further action is needed. They turn yellow when malware has been detected and cleaned. They turn blue when no malware has been detected.

Guard your supply chain

OT zero trust portable security devices can also be used to protect your supply chain. Scan every device before it is deployed. Only allow vendors to access your devices using media that has been scanned and cleansed.

EKANS Ransomware Attack

Ransomware attacks have been a bad brake for automakers recently. In June 2021, EKANS ransomware disrupted auto workers' ability to access computer servers, or to use email or internal systems at an international automobile and motorcycle manufacturer. Production systems around the world were disrupted. No data was stolen, just held for ransom. Forensic analysts were unclear whether the attack was aimed at information technology systems or industrial control systems. Security specialists said that the malware gained access and shut down Microsoft's automated file copy software. Then, it isolated the system by killing all processes that would allow humans to control machines. A variety of different technologies were targeted, including the SCADA systems, virtual machine interfaces, industrial control systems, remote management tools, and network management software. With no access and no automated file copy, even if a security engineer spotted the malware, they could not save any files. After the files were encrypted, the malware sent a ransom message with the title "Fix-Your-Files.Txt". It contained a demand for bitcoin, along with an email address for reply. Following payment, the automaker received a decryption key. Security specialists warned that even though the files were restored, rolling back up to full global operations would take time and have a financial impact.[7]

[7] *Joe Tidy, "Honda's global operations hit by cyber-attack", BBC News (BBC News:2020), https://www.bbc.com/news/technology-52982427 (accessed May 17, 2022).*

EKANS Can't with OT Zero Trust

Threat intelligence gathered by TXOne Networks, along with public threat sources, reveals that EKANS is a prime example of modern ransomware. Machines continue to operate while under an EKANS attack because of a clever design that detects and targets ICS applications. It only sends a ransom note after all the data is encrypted. OT zero trust has protections to stop EKANS before it infects the OT network.

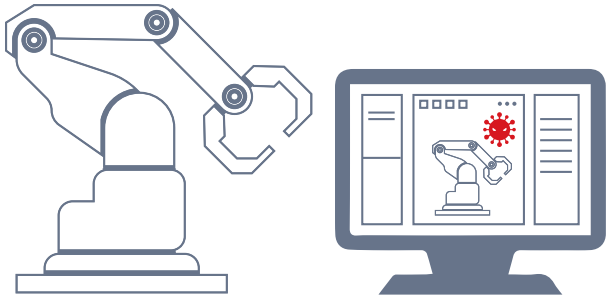
EKANS enters through the IT network. First, before hackers launch EKANS, they must buy credentials from someone on the darkweb or steal them through phishing emails, brute force discovery, or a remote access security hole. They then use these stolen credentials to login and find a weakness in the DMZ (demilitarized zone) from which they can tunnel into the company intranet. Once inside the corporate network, EKANS sets up a stealth communications channel and works toward stealing Active Directory (AD) access credentials. AD is the command center for user authentication and authorization privileges. AD routinely shares security policies so the average OT network trusts AD messages – unless an allow list based on the OT zero trust framework restricts AD privileges, preventing any suspicious or malicious objects from being deployed.

Hackers take advantage of this trust to spread ransomware. They put the malware payload inside a trusted message. It bypasses antivirus checks and shuts down the firewall so that it can spread throughout the network. EKANS malware is smart and can detect when it has infiltrated as many systems as possible. When it detects that the infection is complete, then it encrypts sensitive data before locking out the system and delivering its ransom note. If cybersecurity protections sense mass encryption of data, they can stop it. Once a ransom note has been delivered, it is already too late.

Shift gears and consider how OT zero trust stops or slows down transmission of EKANS ransomware and similar cyber threats. OT zero trust security solutions are capable of detecting and blocking this malware by matching its signature to known malicious code signatures. Ransomware morphs over time and changes signature patterns. OT zero trust leverages threat intelligence to reason about such changes and adapt allowlists and trust lists to prevent new versions of ransomware before they start up. Security is reinforced on key assets through the use of allow lists or portable security scans. Network segmentation and micro-segmentation create zones to slow down or stop the spread of malware.

Segmentation divides the network so you can easily barricade malicious traffic. An OT firewall is deployed like a highway patrol officer stopping intruders from getting in, moving laterally, or gathering information they need to carry out their goals. OT IPSes act like a stoplight monitoring network traffic entering mission-critical devices. They filter protocol traffic and only green-light trustworthy command controls. Just like real traffic lights, OT IPS trust lists adapt to the current flow and type of traffic. Network segments stall attacks and create a window where abnormal and malicious behavior on the network can be identified and potentially infected nodes can be isolated and cleansed before further damage is inflicted.

New Cyber Attack Surface - Digital Twins



One relatively new attack surface in manufacturing is the digital twin. According to Siemens, ***future manufacturing will be driven by the closed-loop digital twin.***[8] A digital twin is a virtual clone. For example, a real robot works in the plant. Its virtual robot twin is online. By creating a constant, closed feedback loop between the real robot and the digital twin, they share the same sensor data and respond to control commands together. The virtual twin has the advantage of being able to gather all sorts of relevant data that can be used to improve the performance of the real robot. The digital twin can gather data from the internet or other sources, including historical performance data. Using machine learning, it can compute the best action to take under the current circumstances and advise the real robot to speed up, slow down, or take precautions if there is a threat.

Digital twins are not limited to robots. You can set up digital twins for some or all of your equipment and gain real-time visibility over the entire plant. When you aggregate data from all your digital twins into a closed-loop simulation, you can discover predictive insights that would have previously been impossible. These insights can be used to increase up-time, improve output efficiency, and lower costs for the entire operation.[9 and 10]

[8] *Siemens Webinar: Virtual commissioning made easy with Process Simulate VC Lite, https://blogs.sw.siemens.com/tecnomatix/virtual-commissioning-made-easy-with-process-simulate-vc-lite/ (accessed May 17, 2022).*

[9] *Revolutionize with intelligent manufacturing, (Siemens Digital Industries Software: 2021) https://static.sw.cdn.siemens.com/siemens-disw-assets/public/6wtgpLabAxpSuB0icuzrsW/en-US/Smart-Manufacturing---White-paper_tcm27-104562.pdf (accessed May 17, 2022).*

[10] *Hazal Şimşek, "Top 5 Use Cases of Digital Twin in Automotive Industry in 2022", AIMultiple Blog (AIMultiple:2022), https://research.aimultiple.com/digital-twin-automotive/ (accessed May 17, 2022).*

Defending Digital Twins

Without virtual guard rails, an evil digital twin can hijack the real robot. Communication between digital twins must be trustworthy. By using OT zero trust allow lists and trust lists, you can confine messages and command controls to only trusted commands for both the real robot and its digital twin. Using the OT zero trust portable security device, you can conduct routine inspections to wipe malware without throttling performance. OT firewalls manage network segmentation that allows you to quarantine both real and digital twins if either of them gets infected.

OT Zero Trust
A New Dimension in Vehicle Quality



Cybersecurity is a new dimension in quality. Drivers and fleet owners must feel secure knowing that their vehicles will behave on and off the road. OT zero trust is a lightweight, supercharged solution for automakers. In the manufacturing plant, you need systems that adapt to what's happening **now**. Interconnected robots or cobots may have been trustworthy yesterday or five minutes ago, but what about **now**? OT zero trust keeps track and grants access privileges based on the current situation and your security policies. The four cornerstones of OT zero trust are the powertrain for cybersecurity.

Inspect



The moment a new asset arrives for the assembly line, inspect it using a portable security device to wipe away any suspicious apps and take an inventory of onboard applications. Automakers and auto parts manufacturers use this portable security device to inspect legacy equipment or devices running old operating systems such as Windows 2000, 7, or XP. Portable security devices are rugged and can be used in harsh environments, even those with temperatures reaching up to 40 degrees Fahrenheit. Portable security devices can also facilitate secure file transfer for data exchange on air-gapped systems. You can use them to perform routine or surprise security inspections without throttling the performance of the device being scanned.

Lock Down



Think of trust lists as stoplights. Stoplights adjust traffic based on current driving conditions. In the automotive manufacturing plant, trustworthiness is repeatedly evaluated based on the current situation. Only trustworthy messages get the green light.

Segment



OT zero trust portable security devices divide your network with safety "cone zones". They also throttle messages through trust lists designed to take the best route based on the current network traffic conditions, the production line situation, and your access control policies. Network segments can be used to sandbox digital twins and protect against corrupt feedback loops.

Reinforce



Barricade systems by stopping both known and unknown attacks. Even zero day threats can be zapped by virtual patching while device manufacturers develop and test security patches. Machine learning reasons about where new threats are ramping up, based on analysis of the latest trends. With OT zero trust, anti-malware computations travel in the fast lane and malware gets caught in speed traps to make sure that production processes always have a green light for safety.

Conclusion

The suite of OT zero trust security devices is built to work 24x7 in any environment. They cover the entire production site so you can view the defense console to see what's going on at any given time. OT zero trust portable security devices are being used to stop supply chain malware from entering the automotive manufacturing plant sector. These security devices are protecting many varieties of legacy and air-gapped equipment. They can even detect cross-platform malware that infects both Windows and Linux devices. Automotive manufacturing plants are using portable security devices for routine and surprise inspections to wipe malware and keep inventory up-to-date. Sometimes data must be hand-carried from robot to robot or device to device. OT zero trust portable security devices provide secure data transport.

OT IPS appliances and OT firewalls adapt to the current circumstances and only allow trustworthy command controls inside network segments. Network segments can be enclaves of robots working together, or a microsegment inside one robot. Security logs provide information to the emergency response teams who complete incident reports. You can always see what's going on by checking the OT defense console. Automakers serious about cybersecurity should take OT zero trust out for a spin.

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